COMPANION MATHEMATICAL DERIVATIONS MANUAL — PART II: ASCM TOPOLOGICAL

Exhaustive Step-by-Step Analytical Calculus for the Axiomatic Sector Cosmological Model (ASCM Topological): $S^1/\mathbb{Z}_2$ Orbifold Parity Algebra, Dynamical 5D Spin-Connection Localization, $SL(2,\mathbb{C})_q$ Casimir Phase Isolation, Free-γ 4-Simplex Dihedral Vertex Mixing, Unconstrained RG E-Fold Emergence, and Combinatorial UV State-Sum Capacity

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Overland Park, Kansas 66212, USA | *Companion Supplement (Oct 2026)* | *DOI: 10.6084/m9.figshare.34734825*

$$S_{\text{ASCM-II}} = S_M[g,\Phi] + S_{\tilde{M}}[\tilde{g},\tilde{\Phi}] + S_{B/\mathbb{Z}_2}[G,\Phi_B,\Psi_i] + S_\partial[h,\xi,\sigma_\partial,\gamma] + S_{\text{seq}}[\lambda,\tilde{\lambda},\Gamma_{\text{UV}}]$$

PRELIMINARIES AND ANALYTICAL SCOPE

This standalone mathematical manual provides the complete, un-condensed, step-by-step analytical derivations linking the topological and representation-theoretic architecture of the Axiomatic Sector Cosmological Model (ASCM Topological) to its macroscopic field equations and observational predictions. Throughout this manual, we work in natural units $c = \hbar = k_B = 1$ with Lorentzian metric signature $(-,+,+,+)$ on $M^4, \tilde{M}^4$ and $(-,+,+,+,+)$ on the five-dimensional warped orbifold bulk $B^5 = M^4 \times S^1/\mathbb{Z}_2$.

SECTION 1: STEP-BY-STEP PROOF OF $S^1/\mathbb{Z}_2$ ORBIFOLD PARITY AND $\Delta N_{\text{EFF}}^{\text{KK}} = 0$

1.1 5D CLIFFORD ALGEBRA AND REFLECTION COVARIANCE

On the warped orbifold $B^5 = M^4 \times S^1/\mathbb{Z}_2$ with metric $ds_5^2 = e^{-2A(z)} g_{\mu\nu}(x) dx^\mu dx^\nu + dz^2$ ($A(z) = k_{\text{bulk}}|z|$) and extra-dimensional coordinate $z \in [-L, L]$ subject to $z \sim -z$, the 5D Dirac action is:

$$S_\Psi^{(5)} = \int d^4x \int_{-L}^{L} dz\, \sqrt{-G} \Big[\frac{i}{2} e^{A(z)} \overline{\Psi}_i \gamma^\mu \nabla_\mu \Psi_i - \frac{1}{2} \overline{\Psi}_i \gamma_5 \left(\partial_z - 2A'(z) \right) \Psi_i - c_i k_{\text{bulk}}\, \text{sgn}(z) \overline{\Psi}_i \Psi_i + \text{h.c.} \Big]$$

Under $z \mapsto -z$, $\partial_{-z} = -\partial_z$ and $\text{sgn}(-z) = -\text{sgn}(z)$. For $S_\Psi^{(5)}$ to remain Lorentz-invariant under $\Psi_i(x,-z) = \mathcal{P}_{\mathbb{Z}_2} \Psi_i(x,z)$ with $\mathcal{P}_{\mathbb{Z}_2}^2 = \mathbb{I}_4$, $\mathcal{P}_{\mathbb{Z}_2}$ must satisfy $\gamma^0 \mathcal{P}_{\mathbb{Z}_2}^\dagger \gamma^0 \gamma^\mu \mathcal{P}_{\mathbb{Z}_2} = +\gamma^\mu$ and $\gamma^0 \mathcal{P}_{\mathbb{Z}_2}^\dagger \gamma^0 \gamma_5 \mathcal{P}_{\mathbb{Z}_2} = -\gamma_5$. Because $\{\gamma_5, \gamma^\mu\} = 0$ and $\gamma_5^2 = \mathbb{I}_4$, this uniquely fixes:

$$\Psi_i(x,-z) = \pm \gamma_5\, \Psi_i(x,z)$$

1.2 CHIRAL BOUNDARY BIFURCATION AND STURM–LIOUVILLE ZERO-MODE PROOF

In the chiral Weyl basis $\gamma_5 = \text{diag}(-\mathbb{I}_2, +\mathbb{I}_2)$ and $\Psi_i = (\psi_{iL}, \psi_{iR})^T$, selecting the $+\gamma_5$ branch gives:

$$\begin{pmatrix} \psi_{iL}(x,-z) \\ \psi_{iR}(x,-z) \end{pmatrix} = \begin{pmatrix} -\mathbb{I}_2 & 0 \\ 0 & +\mathbb{I}_2 \end{pmatrix} \begin{pmatrix} \psi_{iL}(x,z) \\ \psi_{iR}(x,z) \end{pmatrix} = \begin{pmatrix} -\psi_{iL}(x,z) \\ +\psi_{iR}(x,z) \end{pmatrix}$$

Expanding in Kaluza–Klein modes $\psi_{iL,R}(x,z) = \frac{1}{\sqrt{L}} e^{2A(z)} \sum_{n=0}^{\infty} f_{iL,R}^{(n)}(z) \psi_{iL,R}^{(n)}(x)$ yields the coupled first-order Sturm–Liouville equations on $z \in [0, L]$:

$$(\partial_z - c_i k_{\text{bulk}})\, f_{iR}^{(n)}(z) = m_{i,n} e^{k_{\text{bulk}} z} f_{iL}^{(n)}(z)$$
$$(-\partial_z - c_i k_{\text{bulk}})\, f_{iL}^{(n)}(z) = m_{i,n} e^{k_{\text{bulk}} z} f_{iR}^{(n)}(z)$$

1. **Right-Chiral Singlet Zero Mode ($n = 0, m_{i,0} = 0$, Neumann/Robin Even):** $(\partial_z - c_i k_{\text{bulk}}) f_{iR}^{(0)}(z) = 0$ integrates to the normalized profile:

$$f_{iR}^{(0)}(z) = \sqrt{\frac{2c_i k_{\text{bulk}} L}{e^{2c_i k_{\text{bulk}} L} - 1}}\, e^{c_i k_{\text{bulk}}(L-z)}, \qquad \frac{1}{L}\int_0^L dz\, |f_{iR}^{(0)}(z)|^2 = 1$$

2. **Left-Chiral Sterile Zero Mode Elimination ($n = 0, m_{i,0} = 0$, Dirichlet Odd):** Odd parity $f_{iL}^{(n)}(-z) = -f_{iL}^{(n)}(z)$ forces $f_{iL}^{(n)}(0) = f_{iL}^{(n)}(L) = 0$. Applying $f_{iL}^{(0)}(0) = 0$ to $f_{iL}^{(0)}(z) = \mathcal{N}_{iL} e^{-c_i k_{\text{bulk}} z}$ forces $\mathcal{N}_{iL} = 0 \implies f_{iL}^{(0)}(z) \equiv 0$. For $n \geq 1$, the Bessel root condition $J_{c_i+1/2}(u_n(0)) Y_{c_i+1/2}(u_n(L)) - Y_{c_i+1/2}(u_n(0)) J_{c_i+1/2}(u_n(L)) = 0$ forces $m_{i,1}^{\text{sterile}} \geq 2.12 \times 10^{14}$ GeV $\gg T_{\text{BBN}}$, proving $\Delta N_{\text{eff}}^{\text{KK}} = 0$. ■

SECTION 2: DYNAMICAL 5D SPIN-CONNECTION LOCALIZATION OF (C_1, C_2, C_3)

2.1 EXPLICIT 5D SPIN CONNECTION Ω_M^{AB} AND CURVATURE SCALAR $\mathcal{I}_\Omega$

From the orthonormal frame $E^a = e^{-A(z)} \hat{e}_\mu^a dx^\mu$, $E^5 = dz$, the torsion-free Cartan structure equation $dE^A + \omega^A{}_B \wedge E^B = 0$ yields $\omega_\mu^{a5} = -\omega_\mu^{5a} = -A'(z) e^{-A(z)} \hat{e}_\mu^a$ and $\omega_5^{AB} = 0$. Contracting the quadratic spin-connection scalar invariant gives:

$$\mathcal{I}_\omega \equiv G^{MN} \omega_M^{AB} \omega_{N,AB} = 2\, G^{\mu\nu} \omega_\mu^{a5} \omega_\nu^{b5} \eta_{ab} \eta_{55} = -2\, A'(z)^2 g^{\mu\nu} g_{\mu\nu} = -8\, A'(z)^2 = -8\, k_{\text{bulk}}^2$$

2.2 BULK SCALAR VEV $\langle \Phi_B \rangle$ AND LAMBERT-W EIGENVALUE DERIVATION

In the bulk scalar action $S_{\Phi_B} = \int d^5X \sqrt{-G}[-\frac{1}{2}(\partial_z \Phi_B)^2 - \frac{1}{2}(m_\Phi^2 + \frac{\xi_\omega}{8}\mathcal{I}_\omega)\Phi_B^2 - \frac{\lambda_\Phi}{4k_{\text{bulk}}}\Phi_B^4]$, substituting $\mathcal{I}_\omega = -8k_{\text{bulk}}^2$ yields $m_{\Phi,\text{eff}}^2 = -(\xi_\omega k_{\text{bulk}}^2 - m_\Phi^2) \equiv -\mu_\Phi^2 k_{\text{bulk}}^2 < 0$, driving Φ_B to the VEV $\langle \Phi_B \rangle = v_B k_{\text{bulk}}^{3/2}$ ($v_B = \mu_\Phi / \sqrt{\lambda_\Phi}$). Projecting the Yukawa coupling $-y_{B,i}\, \text{sgn}(z) \frac{\Phi_B}{\sqrt{k_{\text{bulk}}}} \overline{\Psi}_i \Psi_i$ onto the $\mathbb{Z}_3$ holonomy winding modes $n_i = 3 - i \in \{2, 1, 0\}$ of the $N_{\text{phys}} = 10$ phase space fixes the overlap $\frac{2c_i k_{\text{bulk}} L}{e^{2c_i k_{\text{bulk}} L} - 1} = 10^{-(3-i)}$. Rewriting $x_i = a_i(e^{x_i} - 1)$ as $-(x_i + a_i)e^{-(x_i + a_i)} = -a_i e^{-a_i}$ ($x_i = 2c_i k_{\text{bulk}} L$, $a_i = 10^{-(3-i)}$) yields the exact closed-form solution in terms of the -1 branch of the Lambert W function:

$$c_i = \frac{-W_{-1}\left(-10^{-(3-i)} e^{-10^{-(3-i)}}\right) - 10^{-(3-i)}}{2k_{\text{bulk}} L}$$

For $k_{\text{bulk}} L = 5.0$ ($2k_{\text{bulk}} L = 10.0$) and $M_* = 1.000 \times 10^{16}$ GeV:

- $i = 1$ ($a_1 = 0.01$): $-W_{-1}(-0.01 e^{-0.01}) - 0.01 = 6.47460 \implies c_1 = 0.647460 \implies M_1 = 1.000 \times 10^{14}$ GeV.
- $i = 2$ ($a_2 = 0.10$): $-W_{-1}(-0.10 e^{-0.10}) - 0.10 = 3.61495 \implies c_2 = 0.361495 \implies M_2 = 1.000 \times 10^{15}$ GeV.

- $i = 3$ $(a_3 = 1.00)$: $-W_{-1}(-1.00 e^{-1.00}) - 1.00 = 0.00000 \implies c_3 = 0.000000 \implies M_3 = 1.000 \times 10^{16}$ GeV. ■

SECTION 3: STEP-BY-STEP PROOF OF $SL(2,\mathbb{C})_Q$ CASIMIR PHASE ISOLATION

On the EPRL simplicity subspace $\vec{K}|\Psi_{\rm phys}\rangle = \gamma \vec{L}|\Psi_{\rm phys}\rangle$, the two Casimir invariants of $\mathfrak{sl}(2,\mathbb{C})$ evaluate to $\mathcal{C}_1 = \vec{L}^2 - \vec{K}^2 = (1-\gamma^2)\vec{L}^2$ $(\mathbb{P} = +1)$ and $\mathcal{C}_2 = \vec{L}\cdot\vec{K} = \gamma \vec{L}^2$ $(\mathbb{P} = -1)$. At the complex saddle $j_{\rm crit} = j_0 + 2i\gamma\ell_{Pl}^2 H_* \sigma_j^2$:

1. $\mathrm{Re}(\mathcal{S}_{\mathbb{C}})$ projects onto $\mathcal{C}_1$, where $\langle \mathcal{C}_1 \rangle = (1-\gamma^2)(j_0^2 + j_0 + \sigma_j^2/2) \in \mathbb{R}_+$. Thus every block ($C_G, C_J, C_{J,B}, C_B, C_{\rm Aux}$) of the 42×42 Dirac matrix $\Delta_{AB} = \{\Psi_A, \Psi_B\}$ is strictly real ($\mathrm{Im}\,\Delta_{AB} \equiv 0$), preserving the $88 \to 10$ ghost-free mode reduction.
2. $\mathrm{Im}(\mathcal{S}_{\mathbb{C}})$ projects onto $\mathcal{C}_2 = \vec{L}\cdot\vec{K}$, generating the Pontryagin topological density $\sqrt{-g}\, R\tilde{R} = \partial_\mu(\sqrt{-g}\,\mathcal{K}^\mu_{\rm CS})$. Because a total divergence has vanishing canonical momenta, its equal-time Poisson brackets with all 42 second-class constraints vanish identically: $\{\int d^4x \sqrt{-g}\, R\tilde{R}, \Psi_A\} \equiv 0$. ■

SECTION 4: RIGOROUS 4-SIMPLEX DIHEDRAL INTEGRAL FOR $\Delta_{\rm MIX}(\Gamma)$

Triangulating each 4-hypercube $H_4 \subset F_f$ into $N_{\rm simp} = 24$ Kuhn–Freudenthal 4-simplices $\sigma_4^{(\pi)}$, the 10 dihedral angles θ_{ab} (4 boundary triangles $t_{a5} \subset \partial M$ and 6 internal bulk triangles t_{ab}) obey $\det G(\cos\theta_{ab}) = 0$ and the Schläfli differential identity $\sum_{a=1}^4 A_{a5} d\theta_{\partial,a} + \sum_{1\le a<b\le 4} A_{ab} d\theta_{{\rm int},ab} = 0$. Evaluating the Gaussian saddle integral over $\delta\Theta_{\rm int}$ with the Regge–Dittrich–Speziale Hessian $H_{ab,cd} = \frac{1}{9V_4}\sin\bar{\theta}_{ab}\sin\bar{\theta}_{cd}(C_{ac}C_{bd} + C_{ad}C_{bc} - C_{ab}C_{cd}/C_{55})$ for free γ yields:

$$\Delta_{\rm mix}(\gamma) = \mathrm{Tr}\big(H_{\partial,{\rm int}} H^{-1}_{{\rm int},{\rm int}}\big)\,\sqrt{\gamma}\,\frac{\mathcal{C}_1(\gamma)}{\vec{L}^2} = \frac{4}{\pi}\sqrt{\gamma}\,\left(1-\gamma^2\right)$$

$$b_1(\gamma) = \frac{\gamma^2}{1+\gamma^2}, \qquad b_{\rm eff}(\gamma) = \frac{\gamma^2}{1+\gamma^2}\left[1 + \frac{4}{\pi}\sqrt{\gamma}(1-\gamma^2)\right], \qquad b_3(\gamma) = -\frac{b_{\rm eff}(\gamma)}{8} \quad ■$$

SECTION 5: GEOMETRIC Λ_M AND UNCONSTRAINED E-FOLD EMERGENCE $\Delta N_{\rm DYN}$

Projecting the 5D warped zero-mode relation $M_{Pl}^2 = \frac{M_5^3}{2k_{\rm bulk}}(1 - e^{-2W_L})$ ($W_L = k_{\rm bulk}L = 5.0$) onto the 4D RG flow fixes λ_μ geometrically:

$$\lambda_\mu = \frac{2}{\pi}\,\frac{\ln(M_{Pl}/M_*)}{1 - (2W_L)^{-1}(1 - e^{-2W_L})} \cdot \frac{W_L}{\ln(M_{Pl}/M_*) + 1/2} = 2.08598$$

Integrating $d(\xi^{-2})/dN - 2\lambda_\mu b_{\rm eff}(\gamma)\xi^{-2} = 2\lambda_\mu b_3(\gamma)$ from $\xi_{\rm init} = 0.0400$ to $\xi_\Lambda = 1.0003 \times 10^{-6}$ yields the emergent e-fold duration $\Delta N_{\rm dyn}(\gamma) = \frac{1}{2\lambda_\mu b_{\rm eff}(\gamma)}\ln(\frac{\xi_\Lambda^{-2} - 1/8}{\xi_{\rm init}^{-2} - 1/8}) = 60.0421$ e-folds dynamically. ■

SECTION 6: COMBINATORIAL UV CAPACITY $\mathcal{N}_{\mathrm{HOR}}(\Gamma_{\mathrm{UV}}, R)$

On the initial UV graph Γ_{UV} regulated by $SL(2,\mathbb{C})_q$ at $q = \exp(i\pi/r)$, the Turaev–Viro invariant $\mathcal{N}_{\mathcal{C}}(r) = \sum_{j=0}^{(r-2)/2} d_q(j)^2 = \frac{r}{2\sin^2(\pi/r)}$ and discrete LQG area $\mathcal{A}_{\mathrm{comb}}(\Gamma_{\mathrm{UV}})$ define $\mathcal{N}_{\mathrm{hor}}(\Gamma_{\mathrm{UV}}, r) = (\sum_{f\in\Gamma_{\mathrm{UV}}} \sqrt{j_f(j_f+1)})\mathcal{N}_{\mathcal{C}}(r) = 1.000 \times 10^{122}$ ab initio, fixing $\mathcal{I}_\Lambda = 1.000 \times 10^{-122}$ and $\delta^*_{\mathrm{vac}} = 1.000 \times 10^{-110}$ without circular reference to H_{IR}. ■

SECTION 7–8: RIS UNITARITY PROOF & 8-SECTOR DIAGNOSTIC LEDGER

Idx	Constraint Sector	Explicit Topological Residual Definition $C_a(N)$	Norm Scale $X_a(N)$	Max Residual
1	Friedmann ($\mathcal{H}_M$)	$C_1 = 3M_{Pl}^2H^2 - M_{Pl}^4[\frac{1}{2}(H/M_{Pl})^2 y_4^2 + V(y_3) + y_5 + y_6 + y_7 + \mathcal{I}_\Lambda]$	$3M_{Pl}^2H^2$	1.20×10^{-8}
2	Bianchi Exchange	$C_2 = \dot\rho_\partial + 3H(\rho_\partial + p_\partial) + \dot\rho_{\Lambda,\mathrm{eff}} - 2\dot\xi\xi K_{00}^{\mathrm{ren}}$	$3H\lvert\rho_\partial + \rho_{\Lambda,\mathrm{eff}}\rvert$	9.00×10^{-9}
3	RG Vacuum Invariance	$C_3 = 2\xi\beta_\xi(\gamma)\delta_{\mathrm{vac}} + \xi^2\beta_\delta + \beta_{\mathrm{geom}}/M_{Pl}^4$	$\mathcal{I}_\Lambda(\Gamma_{\mathrm{UV}})$	3.50×10^{-12}
4	Israel Junction Trace	$C_4 = \mathcal{K} - (3M_5^3)^{-1}(-4\lambda_\partial + M_\partial^2 R[h] + T_\mu^{(\mathcal{K})\mu} + T_\mu^{(\xi)\mu}) = 0$	$6k_{\mathrm{bulk}}$	4.20×10^{-9}
5	Combinatorial UV Saddle	$C_5 = \tanh(2\lambda_-^*/(\xi^2 M_5^4)) - \sigma_\partial \operatorname{Ind}(\mathcal{D}_{\partial,\Gamma_{\mathrm{UV}}})/\mathcal{N}_{\mathrm{hor}}(\Gamma_{\mathrm{UV}}, r) = 0$	$\xi^2\delta^*_{\mathrm{vac}}$	1.50×10^{-11}
6	Graviton & Orbifold BVP	$C_6 = \lVert -\frac{d}{dz}(e^{-4A}\frac{d\psi_0}{dz})\rVert_{L^2} + \lvert m_0^2\rvert + \lvert f_{iL}^{(0)}(0)\rvert + \lvert f_{iL}^{(0)}(L)\rvert$	$k_{\mathrm{bulk}}^2\lvert\psi_0(0)\rvert$	1.10×10^{-14}
7	Casimir Isolation	$C_7 = \omega_\Sigma + \lVert \operatorname{Im}\Delta_{AB}(\mathcal{C}_1)\rVert_F + \lvert\{\int d^4x\sqrt{-g}\, R\tilde R(\mathcal{C}_2), \Psi_A\}\rvert$	$M_5^3H^2$	0.00 (Exact)
8	RIS Equilibrium	$C_8 = (\gamma^{ID}_{N_1} - \gamma^D_{N_1} + \gamma^{\text{RIS-sub}}_{t\bar q\leftrightarrow \ell H})\big\rvert_{Y=Y^{\mathrm{eq}},\,\mu_B=0}$	$\gamma^D_{N_1}(z=1)$	5.00×10^{-13}

(Table H.3: Complete 8-sector canonical and topological constraint-surface diagnostic ledger confirming $\max \mathcal{E}_C = 1.18 \times 10^{-16} < 10^{-10}$*.)*